**RIZQI NAWAF PUTRA ROSYADI**

**103122430010**

**SE0802**



# MODUL 8

# JURNAL

Kerjakan studi kasus pada modul 8 (halaman 87). Sertakan rekaman layar untuk setiap nomor.

## 8.2 Studi Kasus

1. Buatlah desain database untuk suatu perusahaan:
   a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.
   b. Table employee dengan contraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL>
   c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).
2. Analisis table-table dengan compute statistics
3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).
4. Analisis table-table dengan compute statistics
5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.
6. Kemudian tampilkan struktur dari view empvu80.
7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.

```
-- Tabel Department
CREATE TABLE departments (
    department_id INT PRIMARY KEY,
    department_name VARCHAR(50) NOT NULL,
    manager_id INT,
    location_id INT
);

-- Tabel Employee
CREATE TABLE employees (
    employee_id INT PRIMARY KEY,
    last_name VARCHAR(50) NOT NULL,
    email VARCHAR(100),
    salary DECIMAL(10, 2),
    commission_pct DECIMAL(5, 2),
    hire_date DATE NOT NULL,
    department_id INT,
    CONSTRAINT fk_dept FOREIGN KEY (department_id) REFERENCES
departments(department_id)
);

-- Tabel Tambahan: Pemasok (Suppliers)
CREATE TABLE suppliers (
    supplier_id INT PRIMARY KEY,
    supplier_name VARCHAR(100) NOT NULL,
    contact_name VARCHAR(50)
);

-- Tabel Tambahan: Persediaan Barang (Inventory)
```

```
CREATE TABLE inventory (
    item_id INT PRIMARY KEY,
    item_name VARCHAR(100),
    stock_quantity INT,
    supplier_id INT,
    CONSTRAINT fk_supplier FOREIGN KEY (supplier_id) REFERENCES suppliers(supplier_id)
);

ANALYZE TABLE departments COMPUTE STATISTICS;
ANALYZE TABLE employees COMPUTE STATISTICS;
ANALYZE TABLE suppliers COMPUTE STATISTICS;
ANALYZE TABLE inventory COMPUTE STATISTICS;

-- Insert 10 Departments
INSERT INTO departments VALUES (10, 'Administration', 200, 1700);
INSERT INTO departments VALUES (20, 'Marketing', 201, 1800);
INSERT INTO departments VALUES (30, 'Purchasing', 202, 1700);
INSERT INTO departments VALUES (40, 'Human Resources', 203, 2400);
INSERT INTO departments VALUES (50, 'Shipping', 204, 1500);
INSERT INTO departments VALUES (60, 'IT', 205, 1400);
INSERT INTO departments VALUES (70, 'Public Relations', 206, 2700);
INSERT INTO departments VALUES (80, 'Sales', 207, 2500);
INSERT INTO departments VALUES (90, 'Executive', 208, 1700);
INSERT INTO departments VALUES (100, 'Finance', 209, 1700);

-- Insert 10 Employees (Termasuk beberapa di Dept 80 untuk View nanti)
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, department_id) VALUES
```

```
(1, 'King', 'king@mail.com', 5000, 0.1, '2023-01-01', 90),
(2, 'Kochhar', 'koch@mail.com', 4500, NULL, '2023-02-15', 90),
(3, 'Abel', 'abel@mail.com', 3000, 0.2, '2023-03-10', 80),
(4, 'Taylor', 'taylor@mail.com', 3200, 0.2, '2023-04-05', 80),
(5, 'Grant', 'grant@mail.com', 2800, 0.15, '2023-05-12', 80),
(6, 'Whalen', 'whalen@mail.com', 4000, NULL, '2023-06-01', 10),
(7, 'Zlotkey', 'zlot@mail.com', 3100, 0.2, '2023-07-20', 80),
(8, 'Hunold', 'hunold@mail.com', 4200, NULL, '2023-08-14', 60),
(9, 'Ernst', 'ernst@mail.com', 3900, NULL, '2023-09-05', 60),
(10, 'Lorentz', 'lor@mail.com', 2500, 0.1, '2023-10-10', 80);

CREATE VIEW empvu80 AS
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM employees
WHERE department_id = 80;


DESCRIBE empvu80;

-- Hapus index yang sudah ada
DROP INDEX emp_dept_id_idx;

-- Buat ulang index-nya
CREATE INDEX emp_dept_id_idx ON employees(department_id);
```

```
DESCRIBE empvu80;

CREATE INDEX emp_dept_id_idx ON employees(department_id);
```

Script Output

Task completed in 0.038 seconds

```
Name           Null?    Type
-------------- -------- ------------
ID_NUMBER      NOT NULL NUMBER(38)
NAME           NOT NULL VARCHAR2(50)
SALARY                  NUMBER(10,2)
DEPARTMENT_ID           NUMBER(38)
```